MODUL 3

TUGAS PENDAHULUAN

Aturan Tugas Pendahuluan :

1. Buatlah table menggunakan LiveSQL ataupun tools lainnya.
2. **Screenshoot** query dan hasil perubahan pada table.
3. Dilarang plagiarisme.

Soal Tugas Pedahuluan :

1. Buatlah sebuah table mahasiswa dengan atribut: id, nama, tempat lahir, tanggal lahir, nomor hp, email, tinggi badan, berat badan.
2. Lakukan insert data pada table nomor 1 sebanyak 15 rows.
3. Lakukan proses pemrosesan satu table untuk fungsi-fungsi berikut:
   a. Fungsi Karakter dan String (LOWER/UPPER, INITCAP, CONCAT, LENGTH, SUBSTR, LPAD/RPAD, LTRIM, RTRIM, INSTR, REPLACE)
   b. Fungsi Tanggal dan Waktu (MM/MON/MONTH, DD/DY, YYYY/YY/RR, AM (or PM)/HH/HH24, MI/SS, ADD_MONTHS(d,n), SYSDATE, LAST_DAY(d), NEXT_DAY(d,day_name), MONTHS_BETWEEN(d1,d2), ROUND(d,[MONTH| YEAR]))
   c. Fungsi Konversi (TO_NUMBER (char), TO_CHAR (date), TO_CHAR (number), TO_DATE (char))
   d. Fungsi Numerik (ABS(n), MOD (m,n), FLOOR (n), POWER (m,n), ROUND (n[,m]), CEIL (n))

Nama : Rizqi Nawaf Putra Rosyadi

NIM : 103122430010

Kelas : 103122430010

1.

```
CREATE TABLE mahasiswa (
    id NUMBER PRIMARY KEY,
    nama VARCHAR2(100),
    tempat_lahir VARCHAR2(50),
    tanggal_lahir DATE,
    nomor_hp VARCHAR2(15),
    email VARCHAR2(50),
    tinggi_badan NUMBER,
    berat_badan NUMBER
);
```

2. .

```oracle
INSERT ALL
INTO mahasiswa VALUES (1, 'Andi Wijaya', 'Jakarta', TO_DATE('2002-05-15', 'YYYY-MM-DD'), '081234567890', 'andi@mail.com', 170, 65)
INTO mahasiswa VALUES (2, 'Budi Santoso', 'Bandung', TO_DATE('2001-08-22', 'YYYY-MM-DD'), '081234567891', 'budi.s@mail.com', 175, 70)
INTO mahasiswa VALUES (3, 'Citra Lestari', 'Surabaya', TO_DATE('2003-01-10', 'YYYY-MM-DD'), '081234567892', 'citra_l@mail.com', 160, 50)
INTO mahasiswa VALUES (4, 'Dedi Kurniawan', 'Medan', TO_DATE('2002-11-30', 'YYYY-MM-DD'), '081234567893', 'dedik@mail.com', 168, 60)
INTO mahasiswa VALUES (5, 'Eka Putri', 'Palembang', TO_DATE('2003-03-05', 'YYYY-MM-DD'), '081234567894', 'ekap@mail.com', 158, 48)
INTO mahasiswa VALUES (6, 'Fahmi Idris', 'Makassar', TO_DATE('2001-12-12', 'YYYY-MM-DD'), '081234567895', 'fahmi.i@mail.com', 180, 80)
INTO mahasiswa VALUES (7, 'Gita Gutawa', 'Yogyakarta', TO_DATE('2002-07-07', 'YYYY-MM-DD'), '081234567896', 'gita.g@mail.com', 162, 52)
INTO mahasiswa VALUES (8, 'Hadi Saputra', 'Semarang', TO_DATE('2001-04-18', 'YYYY-MM-DD'), '081234567897', 'hadis@mail.com', 172, 68)
INTO mahasiswa VALUES (9, 'Indah Permata', 'Denpasar', TO_DATE('2003-09-25', 'YYYY-MM-DD'), '081234567898', 'indah.p@mail.com', 165, 55)
INTO mahasiswa VALUES (10, 'Joko Susilo', 'Solo', TO_DATE('2002-02-14', 'YYYY-MM-DD'), '081234567899', 'joko_s@mail.com', 178, 75)
INTO mahasiswa VALUES (11, 'Kartika Sari', 'Malang', TO_DATE('2003-06-20', 'YYYY-MM-DD'), '081234567800', 'kartika@mail.com', 160, 50)
INTO mahasiswa VALUES (12, 'Lutfi Hakim', 'Bogor', TO_DATE('2001-10-05', 'YYYY-MM-DD'), '081234567801', 'lutfi.h@mail.com', 174, 72)
INTO mahasiswa VALUES (13, 'Maya Ahmad', 'Aceh', TO_DATE('2002-01-25', 'YYYY-MM-DD'), '081234567802', 'maya_a@mail.com', 159, 49)
INTO mahasiswa VALUES (14, 'Naufal Rizky', 'Padang', TO_DATE('2002-08-14', 'YYYY-MM-DD'), '081234567803', 'naufal@mail.com', 171, 66)
INTO mahasiswa VALUES (15, 'Olivia Rani', 'Manado', TO_DATE('2003-04-12', 'YYYY-MM-DD'), '081234567804', 'olivia@mail.com', 164, 54)
SELECT * FROM dual;
```

3. .

a

```oracle
SELECT
    LOWER(nama) AS nama_kecil,
    UPPER(nama) AS nama_besar,
    INITCAP(tempat_lahir) AS tempat_kapital,
    CONCAT(nama, email) AS gabung,
    LENGTH(nama) AS panjang_nama,
    SUBSTR(nama, 1, 3) AS tiga_huruf_awal,
    LPAD(id, 5, '0') AS id_pad,
    RPAD(nomor_hp, 15, '*') AS hp_sensor,
    LTRIM('   DATA') AS hapus_kiri,
    RTRIM('DATA   ') AS hapus_kanan,
    INSTR(email, '@') AS posisi_at,
    REPLACE(email, 'mail.com', 'gmail.com') AS ganti_domain
FROM mahasiswa;
```

Query Result

All Rows Fetched: 15 in 0.003 seconds

	NAMA_KECIL	NAMA_BESAR	TEMPAT_KAPITAL	GABUNG	PANJANG_NAMA	TIGA_HURUF_AWAL	ID_PAD	HP_SENSOR	HAPUS_KIRI	HAPUS_KANAN	POSISI_AT	GANTI_DOMAIN
3	citra lestari	CITRA LESTARI	Surabaya	Citra Lestaricitra_l@mail.com	13	Cit	00003	081234567892***	DATA	DATA	8	citra_l@gmail.com
4	dedi kurniawan	DEDI KURNIAWAN	Medan	Dedi Kurniawandedik@mail.com	14	Ded	00004	081234567893***	DATA	DATA	6	dedik@gmail.com
5	eka putri	EKA PUTRI	Palembang	Eka Putriekap@mail.com	9	Eka	00005	081234567894***	DATA	DATA	5	ekap@gmail.com
6	fahmi idris	FAHMI IDRIS	Makassar	Fahmi Idrisfahmi.i@mail.com	11	Fah	00006	081234567895***	DATA	DATA	8	fahmi.i@gmail.com
7	gita gutawa	GITA GUTAWA	Yogyakarta	Gita Gutawagita.g@mail.com	11	Git	00007	081234567896***	DATA	DATA	7	gita.g@gmail.com
8	hadi saputra	HADI SAPUTRA	Semarang	Hadi Saputrahadis@mail.com	12	Had	00008	081234567897***	DATA	DATA	6	hadis@gmail.com
9	indah permata	INDAH PERMATA	Denpasar	Indah Permataindah.p@mail.com	13	Ind	00009	081234567898***	DATA	DATA	8	indah.p@gmail.com
10	joko susilo	JOKO SUSILO	Solo	Joko Susilojoko_s@mail.com	11	Jok	00010	081234567899***	DATA	DATA	7	joko_s@gmail.com
11	kartika sari	KARTIKA SARI	Malang	Kartika Sarikartika@mail.com	12	Kar	00011	081234567800***	DATA	DATA	8	kartika@gmail.com
12	lutfi hakim	LUTFI HAKIM	Bogor	Lutfi Hakimlutfi.h@mail.com	11	Lut	00012	081234567801***	DATA	DATA	8	lutfi.h@gmail.com
13	maya ahmad	MAYA AHMAD	Aceh	Maya Ahmadmaya_a@mail.com	10	May	00013	081234567802***	DATA	DATA	7	maya_a@gmail.com
14	naufal rizky	NAUFAL RIZKY	Padang	Naufal Rizkynaufal@mail.com	12	Nau	00014	081234567803***	DATA	DATA	7	naufal@gmail.com
15	olivia rani	OLIVIA RANI	Manado	Olivia Raniolivia@mail.com	11	Oli	00015	081234567804***	DATA	DATA	7	olivia@gmail.com

b

```oracle
SELECT
    TO_CHAR(tanggal_lahir, 'MM/MON/MONTH') AS format_bulan,
    TO_CHAR(tanggal_lahir, 'DD/DY') AS format_hari,
    TO_CHAR(tanggal_lahir, 'YYYY/YY/RR') AS format_tahun,
    TO_CHAR(SYSDATE, 'HH24:MI:SS AM') AS waktu_skrg,
    ADD_MONTHS(tanggal_lahir, 12) AS tahun_depan,
    LAST_DAY(tanggal_lahir) AS akhir_bulan,
    NEXT_DAY(tanggal_lahir, 'MONDAY') AS senin_berikutnya,
    MONTHS_BETWEEN(SYSDATE, tanggal_lahir) AS usia_bulan,
    ROUND(tanggal_lahir, 'YEAR') AS pembulatan_tahun
FROM mahasiswa;
```

Query Result

All Rows Fetched: 15 in 0.004 seconds

	FORMAT_BULAN	FORMAT_HARI	FORMAT_TAHUN	WAKTU_SKRG	TAHUN_DEPAN	AKHIR_BULAN	SENIN_BERIKUTNYA	USIA...	PEMBULATAN_TAHUN
1	05/MAY/MAY	15/WED	2002/02/02	02:42:03 AM	15-05-2003	31-05-2002	20-05-2002	286...	01-01-2002
2	08/AUG/AUGUST	22/WED	2001/01/01	02:42:03 AM	22-08-2002	31-08-2001	27-08-2001	294...	01-01-2002
3	01/JAN/JANUARY	10/FRI	2003/03/03	02:42:03 AM	10-01-2004	31-01-2003	13-01-2003	278...	01-01-2003
4	11/NOV/NOVEMBER	30/SAT	2002/02/02	02:42:03 AM	30-11-2003	30-11-2002	02-12-2002	279...	01-01-2003
5	03/MAR/MARCH	05/WED	2003/03/03	02:42:03 AM	05-03-2004	31-03-2003	10-03-2003	276...	01-01-2003
6	12/DEC/DECEMBER	12/WED	2001/01/01	02:42:03 AM	12-12-2002	31-12-2001	17-12-2001	291...	01-01-2002
7	07/JUL/JULY	07/SUN	2002/02/02	02:42:03 AM	07-07-2003	31-07-2002	08-07-2002	284...	01-01-2003
8	04/APR/APRIL	18/WED	2001/01/01	02:42:03 AM	18-04-2002	30-04-2001	23-04-2001	299...	01-01-2001
9	09/SEP/SEPTEMBER	25/THU	2003/03/03	02:42:03 AM	25-09-2004	30-09-2003	29-09-2003	269...	01-01-2004
10	02/FEB/FEBRUARY	14/THU	2002/02/02	02:42:03 AM	14-02-2003	28-02-2002	18-02-2002	289...	01-01-2002
11	06/JUN/JUNE	20/FRI	2003/03/03	02:42:03 AM	20-06-2004	30-06-2003	23-06-2003	272...	01-01-2003
12	10/OCT/OCTOBER	05/FRI	2001/01/01	02:42:03 AM	05-10-2002	31-10-2001	08-10-2001	293...	01-01-2002
13	01/JAN/JANUARY	25/FRI	2002/02/02	02:42:03 AM	25-01-2003	31-01-2002	28-01-2002	289...	01-01-2002
14	08/AUG/AUGUST	14/WED	2002/02/02	02:42:03 AM	14-08-2003	31-08-2002	19-08-2002	283...	01-01-2003
15	04/APR/APRIL	12/SAT	2003/03/03	02:42:03 AM	12-04-2004	30-04-2003	14-04-2003	275...	01-01-2003

c

```sql
SELECT
    TO_NUMBER('12345') AS ke_angka,
    TO_CHAR(tanggal_lahir, 'DD Month YYYY') AS tgl_ke_teks,
    TO_CHAR(tinggi_badan, '999') || ' cm' AS tinggi_ke_teks,
    TO_DATE('2026-03-19', 'YYYY-MM-DD') AS teks_ke_tgl
FROM mahasiswa;
```

Query Result

SQL | All Rows Fetched: 15 in 0.006 seconds

	KE_ANGKA	TGL_KE_TEKS	TINGGI_KE_TEKS	TEKS_KE_TGL
1	12345	15 May 2002	170 cm	19-03-2026
2	12345	22 August 2001	175 cm	19-03-2026
3	12345	10 January 2003	160 cm	19-03-2026
4	12345	30 November 2002	168 cm	19-03-2026
5	12345	05 March 2003	158 cm	19-03-2026
6	12345	12 December 2001	180 cm	19-03-2026
7	12345	07 July 2002	162 cm	19-03-2026
8	12345	18 April 2001	172 cm	19-03-2026
9	12345	25 September 2003	165 cm	19-03-2026
10	12345	14 February 2002	178 cm	19-03-2026
11	12345	20 June 2003	160 cm	19-03-2026
12	12345	05 October 2001	174 cm	19-03-2026
13	12345	25 January 2002	159 cm	19-03-2026
14	12345	14 August 2002	171 cm	19-03-2026
15	12345	12 April 2003	164 cm	19-03-2026

d

```sql
SELECT
    id,
    tinggi_badan,
    ABS(-170) AS fungsi_abs,
    MOD(id, 3) AS fungsi_mod,
    FLOOR(tinggi_badan / 1.3) AS fungsi_floor,
    CEIL(tinggi_badan / 1.3) AS fungsi_ceil,
    POWER(id, 2) AS fungsi_power,
    ROUND(tinggi_badan / 3, 2) AS fungsi_round
FROM mahasiswa;
```

Query Result

SQL | All Rows Fetched: 15 in 0.004 seconds

	ID	TINGGI_BADAN	FUNGSI_ABS	FUNGSI_MOD	FUNGSI_FLOOR	FUNGSI_CEIL	FUNGSI_POWER	FUNGSI_ROUND
1	1	170	170	1	130	131	1	56,67
2	2	175	170	2	134	135	4	58,33
3	3	160	170	0	123	124	9	53,33
4	4	168	170	1	129	130	16	56
5	5	158	170	2	121	122	25	52,67
6	6	180	170	0	138	139	36	60
7	7	162	170	1	124	125	49	54
8	8	172	170	2	132	133	64	57,33
9	9	165	170	0	126	127	81	55
10	10	178	170	1	136	137	100	59,33
11	11	160	170	2	123	124	121	53,33
12	12	174	170	0	133	134	144	58
13	13	159	170	1	122	123	169	53
14	14	171	170	2	131	132	196	57
15	15	164	170	0	126	127	225	54,67

Semangat!!